

Dhruvit Navadiya

AI Automation Engineer · Agentic Systems · Production AI & Automation Platforms

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AI automation engineer who designs and ships production systems end to end: **multi-agent orchestration**, **multimodal AI pipelines**, and the async backends, job queues and cloud infrastructure that keep them reliable. As a freelancer I am the **primary engineer of an 8-service operations platform (~186K lines of TypeScript and Python)** that runs the warehouse, returns and support operations of a Dutch fulfilment centre, and I built the **billing automation used by Smollan Kenya's finance team**. I architect before I implement, prove correctness with tests, and design for failure recovery rather than demos.

EXPERIENCE

Freelance AI & Automation Engineer — Upwork and direct clients

2025 – Present · Remote

Future Fulfilment B.V. — Warehouse, returns & support operations platform

Mar 2026 – Present · Netherlands

Next.js 16 · TypeScript · Node.js · Python · Firestore · Vertex AI Gemini · Cloud Tasks · Twilio · Railway

649 / 960 commits 8 services 186K source lines 194 API routes 12 prod services ~350 sellers served

- ◆ Primary engineer of the platform the fulfilment centre runs on daily: an internal admin panel, seller portal, AI returns terminal, warehouse TV shift board with 4 cron services, QR clock-in kiosk, carrier-claims email worker, a headless-browser Freshdesk bot and Firebase Functions.
- ◆ Built an **omnichannel support workspace** unifying Gmail (Pub/Sub push sync), WhatsApp (Twilio) and portal tickets, with undoable delayed sends on Cloud Tasks, Gemini reply suggestions and message polish, and a **knowledge base that learns from closed tickets**.
- ◆ Designed an **AI returns terminal**: Gemini reads the shipping label from one photo, a staged resolver matches on tracking code, every spelling of the order number, then customer details, and **HMAC-signed scan tokens** guarantee each label is sent to the model once. Covered by **388 test cases** and a label-reading benchmark.
- ◆ Kept scan and inbox hot paths free of database reads by serving bulk data (a **~14,000-product catalogue**, sellers, open returns) as gzipped Cloud Storage snapshots held in memory with generation checks, and count() aggregations for inbox badges.
- ◆ Integrated the ChannelDock fulfilment API within its hourly rate budget and exact-match filters, and built a white-label deployment of the carrier-claims worker for a second fulfilment company.

Smollan Kenya — Ledgerline, multi-client billing automation

Jul 2026 – Aug 2026 · Kenya

Python · FastAPI · pandas · openpyxl · Next.js 16 · SQLite · Docker · Cloud Run · GitHub Actions

17 client billing configs 0 / 300 cells off in acceptance rebuild 10 months rebuilt to the cent

- ◆ Replaced a hand-maintained monthly Excel billing process with a **deterministic Python pipeline** that joins encrypted payroll (decrypted in memory; passwords never persisted), the HR master, SAGE expenses and IT schedules, then bills each client from **versioned per-client configs**.
- ◆ Writes each month into the client's master workbook as **live COUNTIFS/SUMIFS formulas**, so finance can still audit it in Excel, with pre-commit backups, immutable run snapshots, file hashes and a reconciliation report.
- ◆ Proved correctness three independent ways (a hand-written oracle, the pipeline modules, and a full rebuild of 10 reference months through the HTTP API); all three match finance's workbook **to the cent**.
- ◆ Built a six-stage Next.js operator workflow (upload, join review, blockers and audited overrides, preview, invoiced amount, commit) and shipped it as containers to **Google Cloud Run** through Cloud Build and GitHub Actions.

Nextbase Solutions — AI Automation Engineer

May 2025 – Present · Surat, India

- ◆ Lead the design and delivery of **agentic AI platforms** for media automation, website generation and workflow orchestration, owning each from requirements and architecture through deployment and monitoring.
- ◆ Built **asynchronous Python backends** (AsyncIO, WebSockets, background job queues) that run long-lived AI jobs concurrently and stream live progress to clients.
- ◆ Integrated OpenAI, Gemini on Vertex AI, Fal AI and Whisper with **retries, exponential backoff and schema-validated outputs**, so a failing provider or malformed response never corrupts a job.
- ◆ Engineered modular workflow engines with state tracking, validation loops and automatic error correction; automated cloud asset lifecycles on AWS S3 and Cloudflare R2 and social-media publishing.

Stypix — AI Engineering Intern

Jan 2025 – May 2025 · Remote

- ◆ Prototyped **RAG pipelines** over vector databases and function-calling agents for task execution; built structured prompt pipelines with output-schema validation for deterministic results.
- Coderbabu Infotech — AI & Backend Developer
- Jan 2023 – Apr 2025
- ◆ Built modular **REST APIs in FastAPI and Flask** on PostgreSQL, MongoDB and Redis; shipped reusable automation services for internal tools and supported CI/CD and deployments.

SELECTED PROJECTS

AutoGenix — Agentic website builder

Python · Next.js · Multi-Agent · GitHub API · Vercel

Multi-agent system that turns a plain-language brief into a live site: requirements → design system → architecture → phased code generation → self-correcting validation → GitHub repository → Vercel deployment.

Agentic AI Media Platform

Python · AsyncIO · WebSockets · Vertex AI · S3 / R2 · Pinecone

13-module platform for image, video, caption, transcript, lipsync and document generation: concurrent async jobs, live WebSocket progress, vector-retrieved prompt templates and S3/R2 asset management.

Autonomous AI Video System

Python · LLM APIs · Discord API · Cloud Storage

Clones videos from Instagram or X links via scene detection, or builds them from a script; a JSON scene planner respects model clip limits and keeps characters consistent. Discord-bot control, Instagram auto-publishing.

Ampere — AI support chat & ticketing

Next.js · Node.js · Socket.IO · SQLite · OpenAI

Embeddable chat widget with multi-session recovery, RAG answers grounded in a knowledge base with auto-escalation to humans, and an agent dashboard with live takeover and Discord alerts.

Also: SEO automation platform (Next.js, FastAPI, Gemini) · ComfyUI face-blend workflow (IPAdapter FaceID, SDXL) · real-time warehouse TV dashboard (Firestore) · LinkTrap analytics (BigQuery) · Notion resume dashboard with AI de-duplication.

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, SQL, C/C++
AI & LLM Systems	Multi-agent orchestration, RAG, function calling, structured outputs, LangGraph, OpenAI, Gemini
Generative Media	ComfyUI, SDXL, IPAdapter FaceID, InsightFace, Fal AI, Whisper, image-to-video, FFmpeg
Backend	FastAPI, Flask, AsyncIO, Node.js, WebSockets / Socket.IO, REST API design, job queues
Data	Firestore, PostgreSQL, MongoDB, Redis, SQLite, Pinecone, BigQuery, pandas, openpyxl
Google Cloud	Vertex AI, Cloud Run, Cloud Build, Cloud Tasks, Pub/Sub, Cloud Storage, Firebase Functions
DevOps & Hosting	Docker, GitHub Actions, Railway, Vercel, AWS S3, Cloudflare R2
Integrations	Gmail API, Twilio WhatsApp, GitHub, Notion, Freshdesk, ChannelDock, Playwright / Camoufox
Frontend	Next.js, React, Tailwind CSS, Framer Motion, GSAP
Engineering	pytest, node:test, Jest, golden-file acceptance tests, idempotency, retry and backoff design

EDUCATION

G.H. Patel College of Engineering & Technology — B.Tech, CSE (IoT)

Aug 2021 – May 2025 · Anand, India

CGPA 8.80 / 10, Distinction. Coursework: distributed systems, data structures & algorithms, databases, operating systems, software engineering.

Certifications: Introduction to DevOps · Data Visualization with Python · Introduction to SQL · C++ for C Programmers